AIAssistedCoding

Assignment7.5

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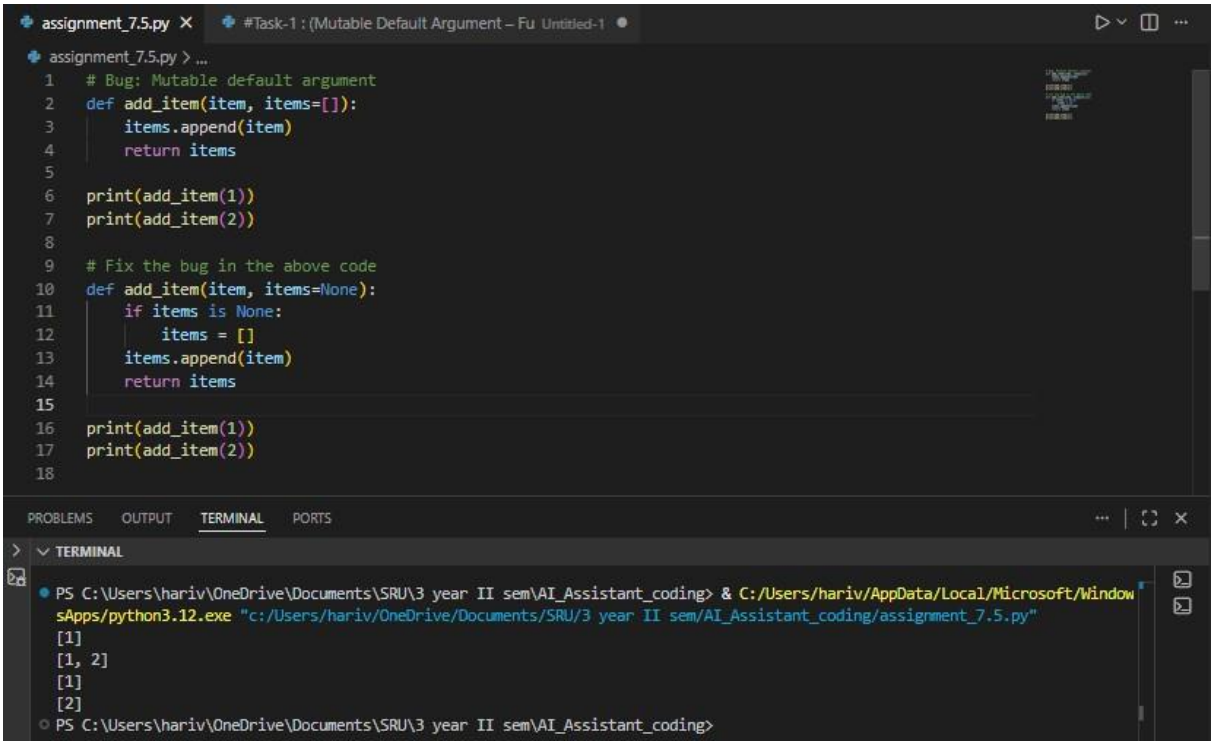
Task 1:

Prompt:

#Bug:Mutabledefaultargument #Fix

the bug in the above code **Code &**

Output:



Explanation:

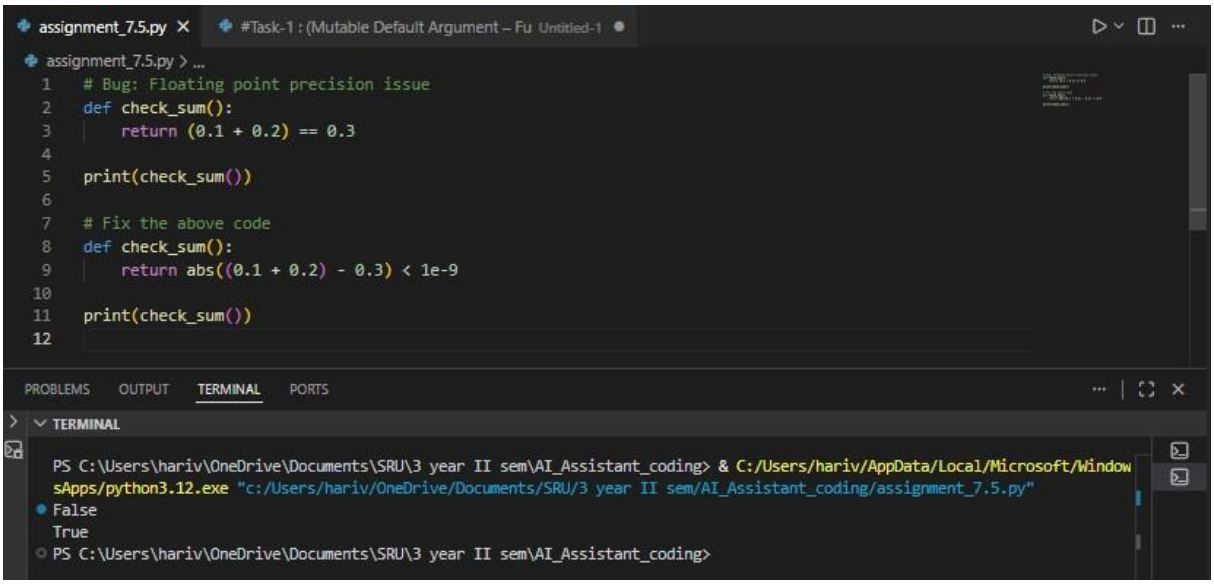
The AI correctly recognized that using a mutable object like a list as a default argument causes the same list to be shared across multiple function calls, leading to unexpected valueaccumulation.Inthecorrectedversion,theAIinitializesthelistinsidethefunction when no argument is passed, ensuring a new list is created each time. This method avoids unwanted side effects and follows recommended best practices for designing Python functions.

Task2:Floating-PointPrecisionError Prompt:

#Bug:Floatingpointprecisionissue #Fix

the above code

Code&Output:



Explanation:

The AI recognized that floating-point values cannot always be compared directly becauseofprecisionissuesintheirbinaryrepresentation.Ratherthanrelyingonexact equality, it recommended comparing the difference between the values within a small acceptable range. This approach makes numerical calculations more accurate and reliable.

Task3:RecursionError–MissingBaseCase

Prompt:

#ThiscodewillcauseaRecursionError:maximumrecursiondepthexceeded incomparison #Fixed

Code:

Code&Output:

```python
# def countdown(n):
#     print(n)
#     return countdown(n - 1)
# countdown(5)
# This code will cause a RecursionError: maximum recursion depth exceeded in comparison

# Fixed code:
def countdown(n):
    if n <= 0:
        print("STOP!")
    else:
        print(n)
        countdown(n - 1)

countdown(5)
```

```
PS C:\Users\hariv\OneDrive\Documents\SRU\3 year II sem\AI_Assistant_coding> & C:/Users/hariv/AppData/Local/Microsoft/WindowsApps/python3.12.exe "c:/Users/hariv/OneDrive/Documents/SRU/3 year II sem/AI_Assistant_coding/assignment_7.5.py"
5
4
3
2
1
STOP!
PS C:\Users\hariv\OneDrive\Documents\SRU\3 year II sem\AI_Assistant_coding>
```

Explanation:

The AI noticed that the recursive function was missing a stopping condition, which led toinfiniterecursionandeventuallyastackoverflow.Byintroducingaproperbasecase, the AI made sure the recursion ends at the right point. This correction highlights how essential exit conditions are when designing recursive algorithms.

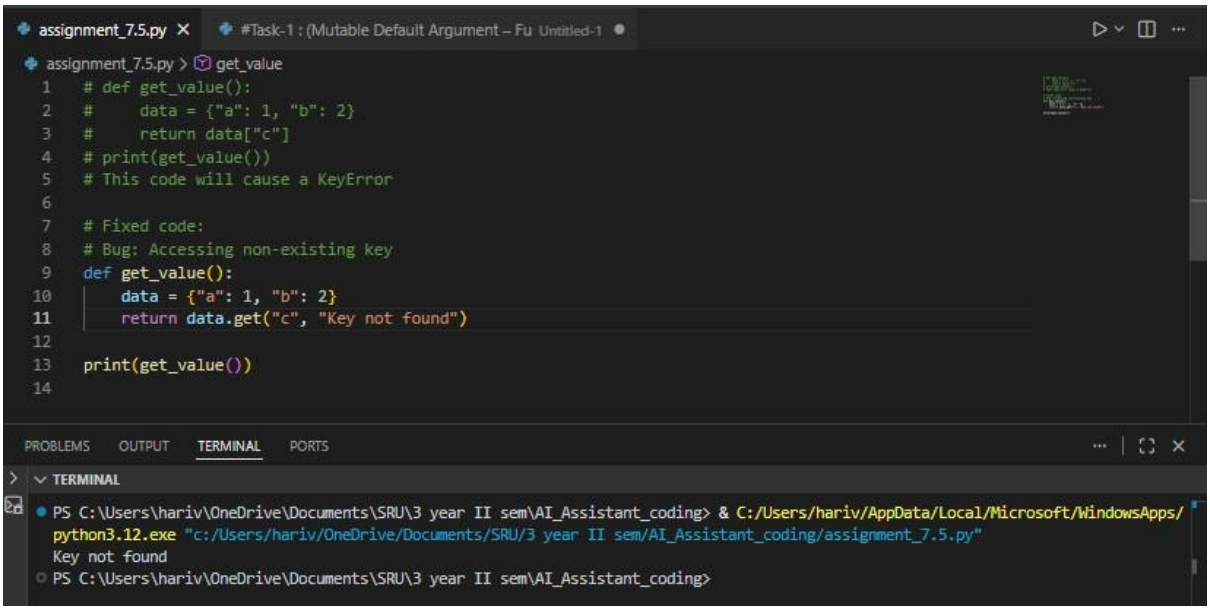
Task4:DictionaryKeyError

Prompt:

#FixedCode:

#Bug:Accessingnon-existingkey

Code&Output:



Explanation:

TheAIunderstoodthattryingtoaccessadictionarykeythatdoesnotexistresultsina **KeyError**. It recommended safer approaches like using the .get() method or properly handling the exception. These alternatives make the program more robust and help avoid unexpected crashes during execution.

Task5:InfiniteLoop–WrongCondition

Prompt:

#Bug:Infiniteloopbecause‘i’isneverincremented. #Fixed

Code:

Code&Output:

```python
# def loop_example():
#     i = 0
#     while i < 5:
#         print(i)
# Bug: Infinite loop because 'i' is never incremented.

# Fixed code:
def loop_example():
    i = 0
    while i < 5:
        print(i)
        i += 1

loop_example()
```

```
PS C:\Users\hariv\OneDrive\Documents\SRU\3 year II sem\AI_Assistant_coding> & C:/Users/hariv/AppData/Local/Microsoft/WindowsApps/python3.12.exe "c:/Users/hariv/OneDrive/Documents/SRU/3 year II sem/AI_Assistant_coding/assignment_7.5.py"
0
1
2
3
4
PS C:\Users\hariv\OneDrive\Documents\SRU\3 year II sem\AI_Assistant_coding>
```

Explanation:

The AI detected that the loop variable was not being updated, which resulted in an infinite loop. By adding the appropriate increment step, the loop now moves steadily towarditsterminationcondition.Thiscorrectionemphasizeshowcrucialitisto update loop control variables properly.

Task6:UnpackingError–WrongVariables

Prompt:

#Bug:Wrong unpacking

#Fix: Changeto correct unpacking

Code&Output:

```python
# a, b = (1, 2, 3)
# Bug: Wrong unpacking
# Fix: Change to correct unpacking
a, b, c = (1, 2, 3)
print(a, b, c)
```

```
PS C:\Users\hariv\OneDrive\Documents\SRU\3 year II sem\AI_Assistant_coding> & C:/Users/hariv/AppData/Local/Microsoft/WindowsApps/python3.12.exe "c:/Users/hariv/OneDrive/Documents/SRU/3 year II sem/AI_Assistant_coding/assignment_7.5.py"
1 2 3
PS C:\Users\hariv\OneDrive\Documents\SRU\3 year II sem\AI_Assistant_coding>
```

Explanation:

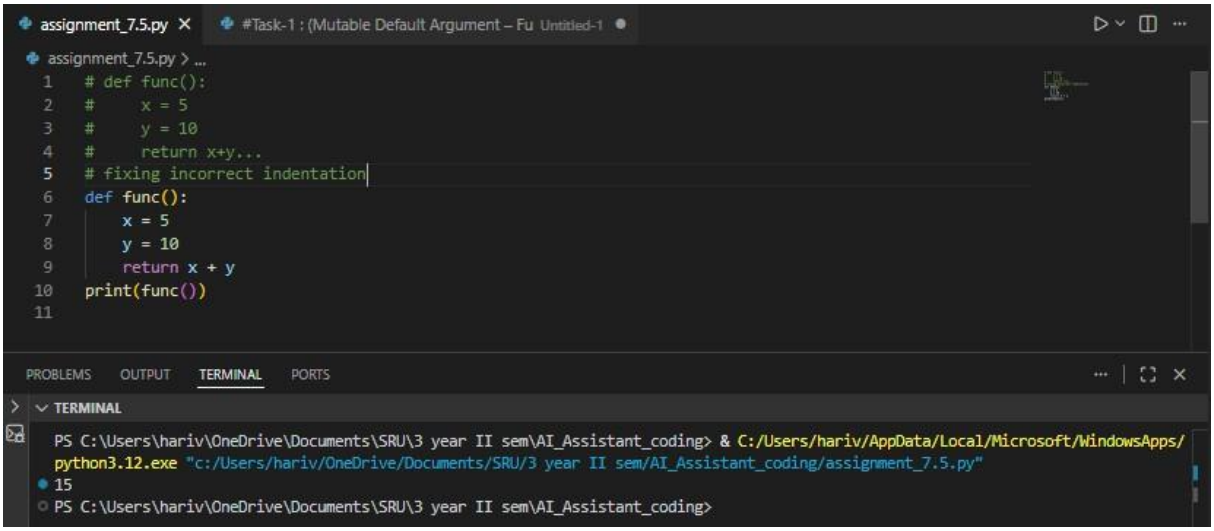
TheAIidentifiedthattherewasamismatchbetweenthenumberofvariablesandthe values being unpacked from the tuple. It suggested fixing this either by adding more variables or by using a placeholder variable (_) for values that are not needed. This approach allows the tuple to be unpacked correctly and avoids runtime errors.

Task7:MixedIndentation–TabsvsSpaces

Prompt:

#Fixingincorrectindentation

Code&Output:



Explanation:

TheAIidentifiedinconsistentindentationasthemainreasonfortheerror.SincePython depends heavily on indentation to define code blocks, the issue caused the program to fail. By fixing the spacing and making the indentation consistent throughout the function, the AI ensured proper execution and also improved the overall readability of the code.

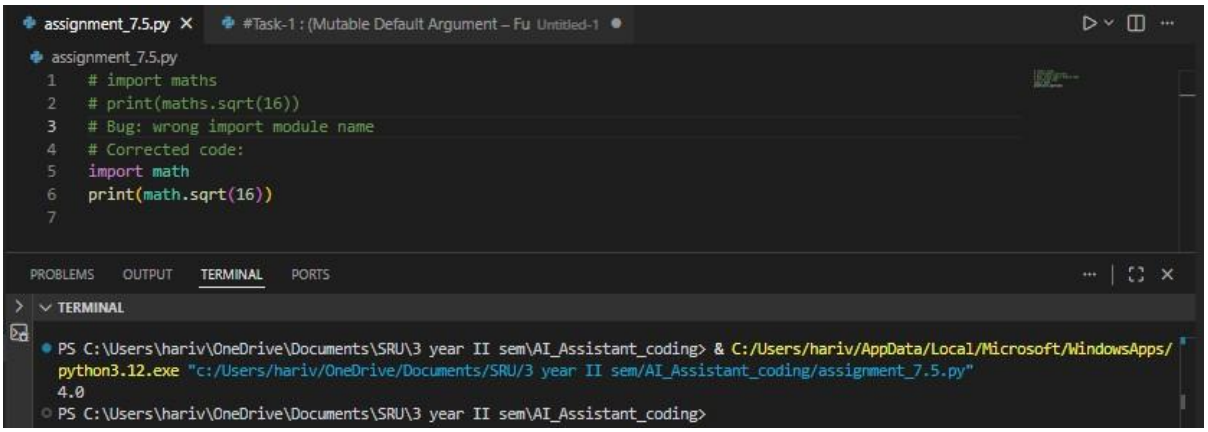
Task8:ImportError–WrongModuleUsage

Prompt:

#Bug:Wrongimportmodulename

#Corrected Code:

Code&Output:



Explanation:

The AI realized that the module name being imported was incorrect. It recommended usingthestandardPython **math**moduleinstead.Thiscorrectionfixestheimporterror and enables the program to access mathematical functions properly.